

BUSINESS REQUIREMENTS DOCUMENT

Business Communication & Call Intelligence Platform

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Table of Contents

Table of Contents.....	2
1 Executive Summary.....	5
2 Business Background	6
3 Problem Statement.....	7
3.1 Lack of Management Visibility.....	7
3.2 Loss of Business Communication History	7
3.3 Lack of Centralized Records	7
3.4 Customer Service Quality	7
3.5 Business Identity	7
4 Business Opportunity.....	8
5 Product Vision.....	9
6 Product Mission	9
7 Product Objectives.....	9
8 Product Scope	10
8.1 MVP Scope	10
9 Out of Scope for Initial MVP	12
10 Target Users.....	13
10.1 Business Owner / CEO	13
10.2 Business Administrator	13
10.3 Agent.....	Error! Bookmark not defined.
10.4 Platform Administrator.....	13
11 User Roles	14
12 Business Onboarding	15
13 KYC Requirements	16
14 Business Account Status	17
15 Company Number Management	18
16 Telephony Architecture	19
17 Recommended Telephony Provider	20
Preferred MVP Provider: Infobip	20
18 Twilio as Alternative	21
19 Agent Calling Model.....	22
20 Recommended MVP Calling Flow	23
21 Call Routing.....	24
22 Call Recording	25
23 Recording Security	26
24 Call Dashboard.....	27

25	Call Details	28
26	Call Statuses.....	29
27	Call Direction.....	30
28	Department Management.....	31
29	Team Management.....	32
30	Agent Availability	33
31	Notifications.....	Error! Bookmark not defined.
32	Billing	Error! Bookmark not defined.
33	Usage Metering.....	36
34	Billing Architecture	37
35	Payment Integration	38
36	Backend Architecture	39
37	Technology Stack	40
38	Why API-First Architecture	41
39	Database	Error! Bookmark not defined.
40	High-Level Entity Relationship	43
41	Multi-Tenancy	44
42	Authentication	Error! Bookmark not defined.
43	Authorization	Error! Bookmark not defined.
44	API Design	47
45	Telephony Webhooks	48
46	Webhook Idempotency	49
47	Background Processing	50
48	Analytics.....	Error! Bookmark not defined.
49	Search and Filtering	52
50	Audit Logging	53
51	Security Requirements.....	54
52	Data Protection and Privacy	55
53	Call Recording Consent	56
54	Recording Retention	57
55	Business Continuity.....	58
56	Reliability Requirements.....	59
57	Monitoring	Error! Bookmark not defined.
58	Deployment	61
59	Recommended Infrastructure	62
60	Functional Requirements.....	63
61	Non-Functional Requirements.....	64
62	MVP Acceptance Criteria	65

63 MVP User Journey.....	66
64 Example Customer Journey	67
65 Business Rules.....	68
66 Subscription Model.....	69
67 Unit Economics	70
68 Market Validation	71
69 Market Validation Success Criteria	72
70 Pilot Program	73
71 Product KPIs.....	74
72 Risks	Error! Bookmark not defined.
73 Telephony Abstraction Layer	76
74 Provider Interface Example	77
75 Critical Product Decision	78
76 Development Phases	79
77 Future Roadmap	80
78 AI Opportunity	81
79 Final MVP Definition	82
80 Final Recommended Architecture	83
81 Immediate Action Plan.....	84
82 BRD Approval Criteria	85
83 Conclusion.....	Error! Bookmark not defined.

1 Executive Summary

The Business Communication & Call Intelligence Platform is a SaaS platform designed to give businesses dedicated corporate telephone numbers through which customer communications can be managed, routed, recorded, and monitored.

The platform addresses a common problem among Nigerian businesses: customer-facing employees frequently use personal mobile numbers to communicate with customers. This creates several business problems:

- ✦ Management has limited visibility into customer interactions.
- ✦ Customer communication history is fragmented.
- ✦ Businesses may lose customer relationships when employees leave.
- ✦ Management cannot easily review how employees handle customers.
- ✦ There is limited evidence for resolving customer disputes.
- ✦ Businesses lack centralized call analytics.
- ✦ There is no standardized corporate communication channel.
- ✦ Customer service quality is difficult to measure.

The proposed platform will provide each verified business with one or more dedicated company numbers. Customers will call the company number rather than an employee's personal number, and the platform will route the call to an appropriate agent while recording and logging the interaction.

Business administrators will be able to view call history, monitor ongoing calls where permitted, review recordings, manage employees and departments, and obtain communication analytics. The platform will operate on a subscription model.

2 Business Background

The original product discussion identified a market problem around businesses using ordinary mobile numbers for customer communication.

The product concept is to provide businesses with a private company number through which customer support and other business calls can be handled and recorded, with administrators able to access recordings through a dashboard. The proposed business model is subscription-based.

The meeting also identified KYC / business verification as an important part of onboarding, with businesses remaining in a pending state until verification is completed, and agreed that market validation should occur before significant development expenditure.

3 Problem Statement

Many SMEs and growing businesses currently depend on personal mobile numbers for customer communication. This creates operational, management, and security problems.

3.1 Lack of Management Visibility

Business owners cannot easily determine:

- ✚ Who spoke with a customer?
- ✚ When the conversation happened.
- ✚ How long the conversation lasted.
- ✚ Whether the customer's issue was resolved.
- ✚ Whether staff behaved professionally.
- ✚ Whether customer complaints are genuine.
- ✚ Whether an employee is mishandling customers.

3.2 Loss of Business Communication History

When an employee leaves a business, customers may continue contacting that employee's personal number, creating dependency on individuals rather than the organization.

3.3 Lack of Centralized Records

Call history is generally stored on individual devices rather than within a company-controlled system.

3.4 Customer Service Quality

Management lacks an effective mechanism for reviewing real customer conversations for:

- ✚ Training
- ✚ Quality assurance
- ✚ Dispute resolution
- ✚ Staff performance
- ✚ Customer service improvement

3.5 Business Identity

Using personal phone numbers reduces the professional identity of the organization. The proposed platform provides a company-controlled communication identity.

4 Business Opportunity

The product can position itself as a lightweight alternative to complex contact-center systems for SMEs. Rather than requiring a business to deploy enterprise call-center infrastructure, the platform provides:

- ✚ Company phone numbers
- ✚ Employee call handling
- ✚ Call recording
- ✚ Call history
- ✚ Department management
- ✚ Basic analytics
- ✚ Administrative monitoring
- ✚ Mobile / web agent access

The product can initially target businesses with approximately 3–50 customer-facing employees. Potential customer segments include:

FinTech

- E-commerce companies
- Schools
- Hospitals and clinics
- Logistics companies
- Real estate companies
- Insurance companies
- Financial institutions
- Retail businesses
- Travel companies
- Professional service firms
- Customer support teams
- SMEs

5 Product Vision

To become the communication infrastructure that gives businesses ownership, visibility, and intelligence over their customer conversations.

6. Product Mission

To replace fragmented personal business calls with a centralized, secure, and measurable company communication platform.

7. Product Objectives

The platform shall:

1. Provide businesses with dedicated company numbers.
2. Allow customers to call those numbers.
3. Route calls to authorized business agents.
4. Record calls where legally and operationally permitted.
5. Store call metadata.
6. Provide administrators with call history.
7. Allow authorized users to play recordings.
8. Allow businesses to organize agents by department.
9. Provide basic communication analytics.
10. Provide secure onboarding and business verification.
11. Support subscription billing.
12. Provide mobile / web communication capabilities for agents.
13. Provide an architecture capable of scaling beyond the MVP.

8 Product Scope

8.1 MVP Scope

The first release shall include:

Business Management

- + Business registration
- + Business profile
- + Business verification
- + KYC submission
- + Account status management
- + Company number assignment

User Management

- + Administrator
- + Agent
- + Department
- + Role management
- + User invitation

Telephony

- + Inbound calls
- + Outbound calls
- + Call routing
- + Call recording
- + Call status tracking
- + Caller identification
- + Call duration
- + Call events

Dashboard

- + Total calls
- + Answered calls
- + Missed calls
- + Ongoing calls
- + Recent calls
- + Call details
- + Recording playback

Agent Interface

- + Receive calls
- + Make calls
- + End calls

- ✚ View recent calls
- ✚ View assigned department
- ✚ Basic call notes

Billing

- ✚ Subscription plan
- ✚ Subscription status
- ✚ Usage tracking
- ✚ Billing history

9 Out of Scope for Initial MVP

The following should not be included in the first version unless market validation demands them:

- ✚ AI call transcription
- ✚ AI sentiment analysis
- ✚ AI coaching
- ✚ WhatsApp calling
- ✚ Video calling
- ✚ Complex contact-center functionality
- ✚ Workforce management
- ✚ Advanced CRM
- ✚ Automated outbound campaigns
- ✚ Predictive dialing
- ✚ Call blasting
- ✚ Advanced speech analytics
- ✚ International number provisioning across many countries
- ✚ Complex IVR trees
- ✚ Full enterprise PBX replacement

These can become Phase 2 / 3 capabilities.

10 Target Users

10.1 Business Owner / CEO

Responsible for the overall organization.

- Visibility
- Reporting
- Call recordings
- Staff monitoring
- Business communication control

10.2 Business Administrator

Manages:

- Employees
- Departments
- Numbers
- Permissions
- Call records
- Subscriptions

10.3 Agent

Handles customer calls, for example:

- Sales representative
- Customer support officer
- Operations officer
- Receptionist

10.4 Platform Administrator

Internal staff responsible for:

- Business verification
- Customer support
- Number allocation
- Platform administration
- Billing issues
- Compliance

11 User Roles

Role	Access
Platform Super Admin	Entire platform
Business Owner	Entire business account
Business Admin	Business management
Department Manager	Department / team management
Agent	Assigned calls and agent functions
Auditor	Read-only access to permitted records

Note: *Permissions must be enforced server-side.*

12 Business Onboarding

Register → Email / Phone Verification → Business Setup → KYC Submission → Pending Verification → Platform Review → Approved → Company Number Assigned → Account Activated

The meeting specifically proposed keeping a business account in a pending state until the business has completed its setup and verification.

13 KYC Requirements

The platform shall support collection of:

Business Information

- Business name
- Business type
- CAC registration number
- Business address
- Business email
- Business phone
- Industry
- Website
- Number of employees

Director / Owner Information

- Full name
- Position
- Contact details
- Identification information where legally required

Documentation

Potential documents:

- CAC certificate
- Business registration documents
- Proof of address
- Director information
- Other documents required by the telephony provider or applicable regulation

Note: The exact KYC requirements shall be finalized after legal and telecom-provider review.

14 Business Account Status

REGISTERED → EMAIL_VERIFIED → KYC_PENDING → UNDER_REVIEW → APPROVED → ACTIVE

Additional states:

REJECTED • SUSPENDED • DEACTIVATED

15 Company Number Management

Each business shall be assigned one or more company numbers, for example:

Field	Example
Business	Acme Logistics
Company Number	+234 XXX XXX XXXX
Departments	Sales, Support, Operations, Accounts

The platform must maintain the relationship:

Business → Phone Number → Routing Configuration → Departments → Agents

16 Telephony Architecture

The MVP should use a third-party CPaaS provider. Recommended architecture:

```
Customer → Company Number → Telephony Provider → Routing Engine → Agent →  
Recording → Webhook → .NET Backend → Database + Object Storage
```

The provider should support:

- Nigerian numbers
- Inbound calls
- Outbound calls
- Call recording
- Webhooks
- Call status events
- Call routing
- Caller ID
- WebRTC or mobile SDK
- Number management
- IVR
- SIP where required

17 Recommended Telephony Provider

Preferred MVP Provider: Infobip

Infobip's Calls API provides application-controlled inbound and outbound calling, call control, call events, and recording management. Its platform also supports WebRTC, mobile SDKs, SIP, and routing capabilities.

Infobip explicitly supports leasing voice numbers for inbound calls and using voice numbers as outbound caller IDs, subject to local regulations and network capabilities.

Important Commercial Validation

Before development begins, the team must obtain:

14. Confirmation of Nigerian inbound voice-number availability.
15. Number types available.
16. Number leasing cost.
17. Setup cost.
18. Inbound per-minute cost.
19. Outbound Nigerian mobile rate.
20. Recording cost.
21. Recording storage cost.
22. WebRTC cost.
23. Number porting capability.
24. Caller-ID rules.
25. Multi-tenant / SaaS resale restrictions.
26. Regulatory requirements.
27. Expected provisioning timeline.

18 Twilio as Alternative

Twilio remains a strong alternative. Twilio currently publishes Nigerian voice termination pricing, including approximately \$0.2303/min for Nigeria and \$0.2349/min for Nigerian mobile destinations on its public pricing page. Call recording is listed at \$0.0025/min plus \$0.0005/min/month for recording storage.

Twilio also supports programmable phone numbers and number management APIs. However, Nigeria-specific number availability and regulatory requirements need to be verified before designing the MVP around Twilio. Twilio's Nigeria guidance requires business documentation for Nigerian mobile numbers, including proof of business identity and address.

Provider	Role
Infobip	Preferred provider — to be investigated first
Twilio	Technical fallback and pricing benchmark

19 Agent Calling Model

The MVP should preferably use:

Option A — WebRTC

The agent opens the web application and receives an in-browser call handled through WebRTC, e.g.:

Field	Example
Event	Incoming call
Customer	+234 XXX XXX XXXX
Department	Sales
Actions	Accept / Reject

Option B — Mobile SDK

A future mobile app integrates directly with the telephony provider's SDK, giving agents a more natural telephone experience.

20 Recommended MVP Calling Flow

Incoming Call

Customer calls company number → Telephony provider → Webhook → .NET backend → Routing engine → Identify available agent → Ring agent → Agent answers → Call recorded → Call ends → Recording finalized → Webhook received → Recording metadata saved → Admin dashboard updated

21 Call Routing

Initial routing can be simple, for example:

Company → Sales Department → Available Agents → Agent 1 / Agent 2 / Agent 3

Routing Strategies

MVP	Future
Round robin	Skills-based routing
First available	Priority routing
Sequential	Working hours
	Department routing
	Agent availability
	IVR selection
	Queue management

22 Call Recording

Every eligible call should produce a call record and an associated recording. The system should store:

- Recording ID
- Call ID
- Provider ID
- Business ID
- Agent ID
- Caller number
- Company number
- Start time
- End time
- Duration
- Recording URL
- File format
- File size
- Retention date

Audio files should be stored in object storage rather than the relational database.

23 Recording Security

Recordings are highly sensitive. The system shall implement:

- Private object storage
- Signed temporary URLs
- Role-based access
- Access logs
- Encryption at rest
- Encryption in transit
- Recording retention policies
- Download restrictions
- Optional watermarking / audit tracking

Note: *An administrator should never receive a permanent public recording URL.*

Instead:

```
GET /api/calls/{id}/recording → Authorize user → Generate signed URL → Return temporary URL
```

24 Call Dashboard

The administrator dashboard shall display:

Summary Cards

- Total Calls
- Answered
- Missed
- Ongoing
- Average Duration

Recent Calls

Caller	Agent	Department	Duration	Status	Time
—	—	—	—	—	—

Filters

- Date
- Agent
- Department
- Status
- Direction
- Duration

25 Call Details

A call details screen should include:

- Caller
- Agent
- Department
- Call direction
- Call status
- Start time
- End time
- Duration
- Recording (with playback control)
- Notes
- Tags

26 Call Statuses

The system shall support:

INITIATED • RINGING • ANSWERED • ONGOING • COMPLETED • MISSED • REJECTED •
FAILED • CANCELLED

27 Call Direction

Calls shall be classified as:

INBOUND • OUTBOUND • INTERNAL

Internal calls may be introduced in a future release.

28 Department Management

Administrators shall be able to create departments, for example:

- Sales
- Support
- Operations
- Accounts
- Logistics
- HR
- Custom departments

Each department shall have:

- Name
- Description
- Manager
- Agents
- Routing rules
- Working hours

29 Team Management

Administrators shall be able to:

- Invite agents
- Edit agents
- Disable agents
- Assign departments
- Change roles
- View agent status
- Remove agents

30 Agent Availability

Agent states:

ONLINE • AVAILABLE • BUSY • ON_CALL • OFFLINE

The routing engine must not route new calls to agents who are unavailable.

31 Notifications

The system may provide:

- Incoming call notification
- Missed call notification
- KYC status notification
- Subscription notification
- Recording availability notification
- Account suspension notification

32 Billing

The platform shall support monthly, bi-annual, and annual plans. Each business shall have:

Subscription → Plan → Usage → Invoice → Payment

33 Usage Metering

Usage should include:

- Call minutes
- Number of active numbers
- Number of agents
- Recording storage
- Optional premium features

Example plan:

Plan	Business Pro
Company numbers	1
Agents	10
Call minutes included	1,000
Recording storage	10 GB

34 Billing Architecture

The billing system should be provider-independent. Paystack and Stripe logic should not be hard-coded directly into business services. Instead, use an abstraction:

```
IBillingProvider → Paystack → Stripe → Future provider
```

This makes the billing layer replaceable.

35 Payment Integration

For Nigeria, Paystack should be considered for subscription payments. The system should support:

- Payment initiation
- Payment verification
- Subscription activation
- Recurring billing
- Failed payments
- Webhooks
- Invoice history
- Payment references

36 Backend Architecture

Recommended stack: ASP.NET Core Web API, organized in layers:

```
API → Application → Domain → Infrastructure → Database
```

Suggested repository structure:

```
src/ → Api / Application / Domain / Infrastructure / Contracts / Shared
```

37 Technology Stack

Backend

- C#
- ASP.NET Core
- Entity Framework Core
- PostgreSQL
- Redis
- Background workers
- REST API
- WebSockets / SignalR where required

Frontend

- React
- TypeScript
- React Query / TanStack Query
- Tailwind CSS or equivalent
- WebRTC
- WebSocket / SignalR

Storage

S3-compatible object storage. Potential options:

- AWS S3
- Cloudflare R2
- Backblaze B2
- DigitalOcean Spaces

38 Why API-First Architecture

The meeting correctly identified that the backend should handle business logic while the frontend consumes APIs. This should remain a core architectural principle, allowing the same backend to support multiple clients:

```
React Web ↔ .NET API → Mobile App → Admin Dashboard → Future Partner APIs →  
Future Integrations
```

39 Database

Recommended database: PostgreSQL. Alternative: SQL Server.

Core entities:

Business • User • Role • Permission • Department • PhoneNumber • Call • Recording •
CallParticipant • RoutingRule • AgentAvailability • Subscription • Plan • Invoice •
Payment • KycApplication • KycDocument • AuditLog • Notification

40 High-Level Entity Relationship

Parent Entity	Related Entities
Business	Users, Departments, Phone Numbers, Calls, Subscription, KYC
User	Role, Department, Calls
Call	Recording

41 Multi-Tenancy

Note: *This is extremely important — the platform is SaaS, so every business's data must be isolated.*

Recommended model: a BusinessId column must exist on all business-owned entities, for example:

Calls table — column	Purpose
Id	Primary key
BusinessId	Tenant identifier
AgentId	Assigned agent
CallerNumber	Caller's number
Duration	Call length

Every query must enforce tenant isolation, e.g. WHERE BusinessId = CurrentBusinessId, and this must never rely solely on frontend filtering.

42 Authentication

Authentication should support:

- Email / password
- Email verification
- Password reset
- Phone verification where required
- Optional MFA
- Session management

JWT / OAuth-compatible authentication can be used.

43 Authorization

The platform shall use role-based access control (RBAC):

Business Owner → Business Admin → Department Manager → Agent

Permissions should include:

calls.read • calls.recordings.read • calls.recordings.download • users.create • users.update
• users.delete • departments.manage • billing.read • billing.manage • business.manage •
kyc.read

44 API Design

Example endpoints:

Method & Path	Purpose
POST /api/auth/register	Register a new account
POST /api/auth/verify	Verify email / phone
POST /api/auth/login	Authenticate a user
GET / PUT /api/business	View / update business profile
POST /api/business/kyc	Submit KYC information
GET /api/business/kyc/status	Check KYC status
GET / POST / PUT /api/users	Manage users
GET / POST /api/departments	Manage departments
GET / POST /api/phone-numbers	Manage company numbers
GET /api/calls, /api/calls/{id}	List / view calls
GET /api/calls/{id}/recording	Retrieve a signed recording URL
POST /api/calls/outbound	Place an outbound call
GET / POST /api/subscription	View / create a subscription
POST /api/webhooks/telephony	Receive telephony provider events
POST /api/webhooks/payment	Receive payment provider events

45 Telephony Webhooks

The telephony provider should notify the backend about events, including:

CALL_RECEIVED • CALL_RINGING • CALL_ANSWERED • CALL_RECORDING_STARTED •
CALL_RECORDING_COMPLETED • CALL_COMPLETED • CALL_FAILED

Webhook processing must be idempotent, authenticated, logged, and retryable.

46 Webhook Idempotency

A provider may send the same event more than once. A unique ProviderEventId should therefore be used to deduplicate events:

```
Does event already exist? → YES → Ignore → NO → Process
```

This prevents duplicate calls and recordings.

47 Background Processing

Background workers should handle:

- Recording processing
- Storage synchronization
- Billing reconciliation
- Email notifications
- Cleanup
- Usage calculations
- Retention deletion
- Analytics aggregation

48 Analytics

MVP Analytics

- Total calls
- Answer rate
- Missed call rate
- Average call duration
- Calls per agent
- Calls per department
- Calls by day
- Calls by hour

Future Analytics

- Customer sentiment
- AI summaries
- Agent quality scores
- Customer satisfaction correlation

49 Search and Filtering

Administrators should be able to search by caller number, agent, department, date, and call ID. Filters should be server-side.

50 Audit Logging

The platform must maintain audit logs for sensitive activities, for example:

- Admin played a recording
- Admin downloaded a recording
- Agent changed department
- Admin invited a user
- Admin removed a user
- KYC document accessed
- Subscription changed
- Phone number assigned

Each audit log entry should capture:

Id • BusinessId • UserId • Action • ResourceType • ResourceId • IP • UserAgent •
Timestamp • Metadata

51 Security Requirements

The system shall implement:

- TLS
- Secure authentication
- RBAC
- Tenant isolation
- Encryption at rest
- Secure secrets management
- API rate limiting
- Input validation
- SQL injection protection
- XSS protection
- CSRF protection where applicable
- Secure webhook verification
- Signed recording URLs
- Audit logging

52 Data Protection and Privacy

Call recordings and caller information should be treated as sensitive personal data. The Nigeria Data Protection Act requires personal-data processing to have a lawful basis and requires transparency regarding purposes, recipients, rights, and retention periods.

The product should therefore incorporate:

- Privacy notice
- Recording notification
- Defined lawful basis
- Retention policy
- Access controls
- Data deletion procedures
- Data-subject request processes
- Data breach procedures

Note: *The exact legal basis for call recording should be reviewed by Nigerian privacy counsel before production.*

53 Call Recording Consent

The safest product design should not silently record calls. The platform should support a configurable announcement such as:

“This call may be recorded for quality and training purposes.”

The announcement should be configurable per business, for example:

Setting	Example Value
Recording	ON
Announcement	ON, played before recording starts
Language	English
Retention	90 days

Note: For legal certainty, the business should obtain professional legal advice on the appropriate recording / consent model for its use case and any cross-border callers.

54 Recording Retention

Default retention options could include:

30 days • 90 days • 180 days • 365 days • Custom

The system should automatically delete expired recordings. Metadata may be retained longer where appropriate.

55 Business Continuity

Telephony failure must not completely destroy the customer communication channel. Future capabilities:

- Failover number
- Secondary routing
- Provider fallback
- SIP fallback
- Call forwarding
- Emergency routing

56 Reliability Requirements

Milestone	Target Availability
MVP	99.5%+
Future	99.9%+

Critical systems that must be monitored: authentication, telephony webhooks, call routing, recording metadata, and billing.

57 Monitoring

Implement:

- Application logs
- Error tracking
- API metrics
- Telephony metrics
- Webhook monitoring
- Database monitoring
- Infrastructure monitoring

Alerts should cover:

- Failed calls
- Webhook failures
- Recording failures
- High latency
- Database errors
- Storage failures
- Payment failures

58 Deployment

Initial architecture components: .NET API, React application, PostgreSQL, Redis, and a background worker, hosted on a cloud VPS.

The original meeting discussed starting inexpensively and migrating to a VPS as usage grows. For the actual telephony MVP, traditional shared cPanel hosting for the backend is not recommended.

A small VPS / cloud environment is more appropriate because the platform will need:

- Webhooks
- Background workers
- HTTPS
- WebSockets
- Redis
- Monitoring
- Persistent processes
- Secure networking

59 Recommended Infrastructure

Layer	Recommendation
Frontend	Cloudflare / Vercel
Backend	Render / Railway / VPS
Database	Managed PostgreSQL
Redis	Managed Redis
Storage	S3-compatible object storage
Telephony	Infobip
Payments	Paystack

This gives the team a relatively simple starting point.

60 Functional Requirements

ID	Requirement
FR-001	Registration — the system shall allow a business representative to create an account.
FR-002	Verification — the system shall verify the user's email and/or phone.
FR-003	KYC — the system shall collect required business verification information.
FR-004	Approval — only approved businesses may activate telephony services.
FR-005	Number Assignment — the system shall assign a dedicated number to an approved business.
FR-006	Call Reception — the system shall receive inbound calls.
FR-007	Call Routing — the system shall route calls to eligible agents.
FR-008	Recording — the system shall record eligible calls.
FR-009	Call Logging — the system shall record call metadata.
FR-010	Playback — authorized administrators shall be able to play recordings.
FR-011	Team Management — administrators shall manage team members.
FR-012	Departments — administrators shall manage departments.
FR-013	Billing — the system shall manage subscriptions and payments.
FR-014	Analytics — the system shall provide call analytics.

61 Non-Functional Requirements

Category	Requirement
Performance	API responses should generally target <500ms for normal database operations. Telephony events must be processed asynchronously where possible.
Scalability	The system should support horizontal scaling.
Security	All sensitive communication must use HTTPS/TLS.
Availability	Target 99.5% MVP availability.
Maintainability	Code must follow established .NET architectural patterns.
Observability	All critical workflows must be traceable.

62 MVP Acceptance Criteria

The MVP is considered successful when the following are achieved:

Business can

- Register
- Verify email
- Submit KYC
- Become approved
- Receive a company number
- Add agents
- Create departments
- Configure routing

Customer can

- Call the company number
- Reach an appropriate agent

Agent can

- Receive a call
- Answer it
- End it
- Make an outbound call
- See call history

Administrator can

- View calls
- Filter calls
- View call details
- Play recordings
- View agents
- View departments
- Monitor usage

Platform can

- Receive telephony webhooks
- Store call metadata
- Store recordings securely
- Enforce tenant isolation
- Track billing usage

63 MVP User Journey

Business Owner → Register → Verify Email → Complete Business Profile → Submit
KYC → Account = Pending → Platform Review → Approved → Choose Subscription →
Payment → Company Number Assigned → Create Departments → Invite Agents →
Configure Routing → Go Live

64 Example Customer Journey

Customer calls company number → Platform receives call → Identifies department → Checks available agents → Rings agent → Agent answers → Recording begins → Conversation → Call ends → Recording stored → Admin sees call

65 Business Rules

ID	Rule
BR-001	Unverified businesses cannot activate telephony services.
BR-002	Only authorized administrators can access recordings.
BR-003	Agents can only access calls permitted by their role.
BR-004	A disabled agent cannot receive new calls.
BR-005	Expired subscriptions may restrict outbound calling.
BR-006	Call recordings must follow the configured retention period.
BR-007	Every recording access must be auditable.
BR-008	Every telephony event must be traceable to a business.
BR-009	A phone number can only belong to one active business tenant at a time.

66 Subscription Model

Plan	Numbers	Agents	Minutes	Recording	Analytics
Starter	1	5	Limited	Basic	Basic
Business	2	20	Higher allowance	Included	Advanced
Enterprise	Multiple	Unlimited / large	Custom	Custom retention	Advanced reporting + dedicated support

Note: Exact pricing must be determined after market validation and telephony cost analysis.

67 Unit Economics

This is one of the most important areas before launch. For every customer, gross margin is calculated as:

Subscription Revenue → - Telephony Minutes → - Number Cost → - Recording Cost →
- Storage → - Payment Processing → - Infrastructure → = Gross Margin

Indicative provider pricing (currencies differ by provider and should be normalized to a single currency once formal quotes are obtained):

Provider	Voice (outbound, Nigeria)	Recording	Storage
Twilio	~\$0.23/minute	\$0.0025/minute	\$0.0005/minute/month
Infobip	Quote required (destination-dependent)	€0.0021/minute	€0.000048/MB/month

Note: Do not price the SaaS subscription until formal costs are known. The team should obtain a formal Infobip quote before deciding pricing.

68 Market Validation

Before full development, interview at least 10–20 businesses. Suggested questions:

- 28. How do you currently handle customer calls?
- 29. Do employees use personal numbers?
- 30. Do you have a company number?
- 31. How many customer calls do you receive monthly?
- 32. Do you record calls?
- 33. How do you handle customer disputes?
- 34. Have you experienced staff / customer communication problems?
- 35. Would you pay for a dedicated company number?
- 36. Would call recording be valuable?
- 37. Would you pay monthly for this service?
- 38. What would make you trust the platform?
- 39. What price would feel reasonable?

69 Market Validation Success Criteria

Proceed aggressively if:

- At least 50% of interviewed businesses identify the problem.
- At least 30% demonstrate willingness to pay.
- At least 5 businesses agree to pilot.
- At least 2 businesses agree to pay for a pilot.

Note: *The strongest validation is payment, not positive feedback.*

70 Pilot Program

Parameter	Target
Businesses recruited	5
Agents per business	5–20
Duration	30–60 days

Measure:

- Calls
- Minutes
- Missed calls
- Recording usage
- Admin dashboard usage
- Agent usage
- Customer complaints
- Retention
- Willingness to pay

71 Product KPIs

Acquisition

- Number of businesses registered
- KYC completion rate
- Approval rate

Activation

- Businesses receiving first call
- Time to first call
- Agents activated

Engagement

- Calls per business
- Minutes per business
- Recording playback
- Dashboard sessions

Revenue

- MRR
- ARPU
- Churn
- Gross margin
- Cost per business

Operational

- Call answer rate
- Call failure rate
- Recording failure rate
- Webhook failure rate

72 Risks

Risk	Description	Mitigation
1 — Telephony Cost	High Nigerian voice termination costs could make the product unprofitable.	Obtain volume pricing and design plans around usage.
2 — Number Availability	The entire product depends on suitable Nigerian business numbers.	Validate provider number inventory before development.
3 — Regulatory Compliance	Voice recordings contain personal data.	Privacy-by-design, lawful basis, notices, retention, and legal review.
4 — Provider Dependency	Relying entirely on one CPaaS creates vendor lock-in.	Create a Telephony Abstraction Layer.

73 Telephony Abstraction Layer

This is highly recommended. Instead of embedding Infobip-specific code throughout the codebase:

```
CallsController → CallService → ITelephonyProvider → InfobipTelephonyProvider
```

This allows additional providers to be added later:

```
ITelephonyProvider → InfobipProvider → TwilioProvider → OtherProvider
```

This makes it possible to switch providers without rebuilding the entire platform.

74 Provider Interface Example

Conceptually, `ITelephonyProvider` should expose:

`ProvisionNumber()` • `ReleaseNumber()` • `CreateCall()` • `EndCall()` • `ConfigureRouting()` •
`StartRecording()` • `StopRecording()` • `GetCall()` • `GetRecording()` • `ValidateWebhook()`

This should be part of the architecture from Day 1.

75 Critical Product Decision

The team should NOT frame the build as “building a Twilio application” or “building an Infobip application.” Instead, the framing should be:

“We are building a business communication platform that currently uses Infobip as its telephony infrastructure provider.”

That distinction is strategically important.

76 Development Phases

Phase	Duration	Deliverables
0 — Validation	1–2 weeks	Market interviews; provider evaluation; Infobip commercial discussion; Twilio comparison; regulatory review; final MVP scope.
1 — Foundation	1–2 weeks	Repository; .NET API; database; authentication; React application; CI/CD; environments.
2 — Business Management	1–2 weeks	Registration; verification; business profile; KYC; admin users; departments.
3 — Telephony	2–4 weeks	Provider integration; number provisioning; inbound/outbound calls; routing; webhooks; recording.
4 — Dashboard	2–3 weeks	Overview; call logs; call details; recording playback; agent management; departments; analytics.
5 — Billing	1–2 weeks	Plans; Paystack; subscription; usage; invoices.
6 — Pilot	4–8 weeks	5 pilot businesses; monitoring; bug fixes; usage analysis; pricing validation.

77 Future Roadmap

Version	Focus
1.1	Better routing, agent availability, call notes, tags, search, export.
1.2	Mobile application, push notifications, advanced analytics.
2.0	AI transcription, AI summaries, sentiment analysis, agent scoring, call quality intelligence.
3.0	CRM integration, WhatsApp integration, email integration, omnichannel communication.

78 AI Opportunity

Once sufficient recordings exist, the platform can evolve from a Call Recording Platform into a Call Intelligence Platform:

Recording → Transcription → AI Analysis → Summary → Sentiment → Customer
Intent → Agent Score → Action Items

This is potentially a much stronger long-term differentiator than simply recording calls.

79 Final MVP Definition

The MVP should answer one question:

“Can a business replace its employees' personal customer-facing numbers with a controlled company communication system and gain centralized visibility into those calls?”

If the answer is yes, the MVP is successful. The first version does not need to become a full contact-center platform.

80 Final Recommended Architecture

```
Customer → Infobip – Voice Infrastructure → ASP.NET Core API  
ASP.NET Core API → PostgreSQL → Redis → Workers → Object Storage (Recordings)  
React + TypeScript → Admin Dashboard → Agent Interface
```

81 Immediate Action Plan

Note: The team should not start full backend development yet.

Step	Action
Step 1	Finish the Figma screens.
Step 2	Freeze the MVP scope.
Step 3	Contact Infobip with a clear use-case description (see below).
Step 4	Ask Twilio exactly the same questions.
Step 5	Build a small telephony proof-of-concept, not the whole product.
Step 6	Only after this works should you commit to a telephony provider.
Step 7	Then build the complete MVP.

Suggested outreach message for Step 3:

"We are building a Nigerian multi-tenant SaaS platform where each verified business customer receives a dedicated Nigerian voice number. Customers call that number, calls are routed to business agents through WebRTC/mobile applications, calls are recorded, and recordings are exposed through our SaaS dashboard. Do you support this use case and can you provide Nigerian voice numbers for this model?"

Request the following from both providers:

- Number availability
- Inbound rate
- Outbound rate
- Number rental
- Setup fees
- Recording fees
- Storage fees
- WebRTC pricing
- Mobile SDK pricing
- Number porting
- Caller ID rules
- Regulatory / KYC requirements
- Multi-tenant SaaS terms
- Volume discounts

The proof-of-concept should prove:

Nigerian number → Incoming call → .NET webhook → Route to agent → Agent answers
→ Record → Recording webhook → Store recording → Play recording from React

82 BRD Approval Criteria

The BRD should be considered approved when the following are confirmed:

☐	Target customer defined
☐	MVP scope approved
☐	Figma approved
☐	Telephony provider selected
☐	Nigerian number availability confirmed
☐	Pricing confirmed
☐	Regulatory requirements reviewed
☐	KYC requirements defined
☐	Subscription model defined
☐	API architecture approved
☐	Database model approved
☐	Security requirements approved
☐	Pilot businesses identified
☐	Development roadmap approved

83 Conclusion

The opportunity is not simply to build another VoIP application. The stronger proposition is:

A business communication control and intelligence platform for African businesses.

The first version provides:

```
Company Number → Calls → Agents → Recording → Dashboard
```

The second version becomes:

```
Calls → Transcription → AI → Insights → Staff Performance → Business  
Intelligence
```

That gives the product a much stronger long-term direction. For the MVP, the recommended approach is:

- React + TypeScript
- ASP.NET Core
- PostgreSQL
- Redis
- Object Storage
- Infobip (subject to Nigerian number / commercial validation)
- Paystack
- WebRTC / Mobile SDK for agents

The most important immediate task is therefore not coding — it is proving that the Nigerian telephony infrastructure, regulatory requirements, and unit economics work for the SaaS model.